

Akash Yaduwanshi

Indore, India | +91-7772074181 | aakashyaduwanshi0470@gmail.com

GitHub: github.com/unshakensoul17 | LinkedIn : <https://www.linkedin.com/in/akash-yaduwanshi-902a3b352> | Portfolio : portfoliowebsite-1nq9.vercel.app/

Summary

AI & Machine Learning Engineer with experience building end-to-end intelligent systems, including NLP pipelines, transformer-based models, and real-time AI applications. Skilled in deep learning, scalable system design, and deploying production-ready AI solutions using PyTorch and FastAPI.

Education

Bachelor of Computer Applications (BCA) Holkar Science College, Indore | Expected 2027

- **CGPA:** 8.60 / 10.0
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Technical Skills

- **Programming:** Python , SQL, TypeScript
 - **AI / Machine Learning:** PyTorch, TensorFlow, Transformers, NLP, LLMs, SentenceTransformers
 - **Systems & Deployment:** FastAPI, Docker, Redis, PostgreSQL, SQLite
 - **Data Tools:** Pandas, NumPy, Matplotlib, Seaborn, PCA, Clustering
 - **Tool:** Git, Github, Claude code, Antigravity, Vs code
 - **Core Areas:** Deep Learning, NLP, Transformers, Representation Learning
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Projects

Sentinel Flow — AI Code Intelligence System

Link : [Marketplace](#)

(TypeScript, React, SQLite, Tree-sitter, AI APIs)

- Developed an AI-powered code intelligence system that transforms codebases into knowledge graphs for real-time structural and dependency analysis.
- Designed a dual-path AI routing system (Groq, Gemini, Bedrock) delivering <300ms responses for quick insights and deep architectural analysis within the editor.
- Built graph-based visualizations using React and ELK.js to detect technical debt and map change impact (blast radius).
- Optimized AST parsing using worker-thread isolation, reducing update latency to <500ms while maintaining UI responsiveness.

AI Resume Analyzer — NLP-Based Scoring System

Link: [Github](#)

Python, PyTorch, Sentence Transformers, FastAPI)

- Developed an end-to-end ML system combining semantic embeddings with engineered features (ATS metrics, skills, experience) to produce category-wise relevance scores.

- Designed a multi-head attention scoring model with multi-task outputs and deployed the service via FastAPI for real-time evaluation.
- Implemented robust document parsing for PDF/DOCX/TXT to extract structured data for downstream analysis.
- **Leveraged Sentence Transformer embeddings for semantic similarity, improving contextual matching beyond keyword-based approaches.**

SmartCart Customer Segmentation — Unsupervised ML Pipeline

Link : [Github repo](#)

(Python, Scikit-learn, PCA, K-Means)

- Engineered a robust unsupervised pipeline identifying 4 actionable customer segments for targeted marketing insights.
- Applied PCA for dimensionality reduction and validated cluster quality using silhouette scores and business-focused profiling.

Compositional Reasoning in Transformers via Architectural Bottlenecks

Link : [Report](#)

(Transformers, Deep Learning, PyTorch, SCAN Benchmark)

- Implemented slot-based reasoning bottlenecks in small Transformers to induce compositional generalization, improving mean accuracy from **~0.55** to **~0.71** on the SCAN benchmark while keeping model size constant.
- Investigated the emergence of reasoning as a capacity-dependent phase transition, identifying instability across random initializations.
- Evaluated hybrid quantum-classical bottlenecks, demonstrating the performance impact of lack of structured inductive bias in naive quantum feature maps.

Certifications & Achievements

- **Python Programming Certification** — Programmers Point, Indore
 - Built and published multiple AI projects on GitHub; comfortable with end-to-end ML pipeline and deployment.
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